

# A Counterexample to Compact Refinement for the Approximate Unique Extension Property

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**Status.** Candidate counterexample, likely valid. This packet answers negatively the compact-refinement question raised after Theorem 3.6 of Thompson’s paper [1].

## 1 Source and Question

The source paper studies the approximate unique extension property (AUEP) for unital completely positive extensions of representations of operator spaces. After proving that approximate unitary equivalence to a representation with the unique extension property implies AUEP, Thompson compares the result with Voiculescu’s theorem and asks whether, under separability assumptions, the approximating errors

$$\pi(t) - u_\beta^* \psi(t) u_\beta$$

can always be chosen compact for every  $t$ .

The relevant source passage is on pages 11–12 of [1].

We remark that one form of Voiculescu’s Theorem states that, whenever  $\pi$  and  $\sigma$  are approximately unitarily equivalent  $*$ -representations of a separable  $C^*$ -algebra  $\mathfrak{A}$  on a separable Hilbert space, then the limiting difference  $\sigma(t) - u_n^* \pi(t) u_n$  can be made compact for every  $t \in \mathfrak{A}$  and every  $n \in \mathbb{N}$  [45, Theorem 1.5]. A similar statement has been obtained for tuples of separable operators [14, Theorem 5.2]. Thus, one may wonder whether the difference in our setting, given by  $\pi(t) - u_\beta^* \psi(t) u_\beta$ , can be made compact under separability assumptions. However, we do not know if

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such a refinement is possible in our context due to the application of compactness in Proposition 3.5.

Since the usual unique extension property is not preserved under approximate

## 2 Idea of the Proof

The compactness refinement is too strong because compact perturbation is visible in the Calkin algebra. If a unitary  $u$  made  $\pi(t) - u^* \psi(t) u$  compact for every  $t$ , then the Calkin image of  $\psi$  would be unitarily conjugate to the Calkin image of the representation  $\pi$ . Thus the Calkin image of  $\psi$  would have to be multiplicative.

Now start with an AUEP representation that is not UEP. It has a nontrivial UCP extension, hence a nonmultiplicative one. Countably amplify both maps. The multiplicative defect becomes an infinite diagonal repeat of a nonzero operator, so it is not compact. This contradicts the Calkin

multiplicativity forced by compact errors. Thompson's Cuntz-algebra example supplies the required separable AUEP-but-not-UEP representation.

### 3 A General Amplification Obstruction

Let  $\mathcal{Q}(\mathcal{H}) = \mathcal{B}(\mathcal{H})/\mathcal{K}(\mathcal{H})$  denote the Calkin algebra and let  $q : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{Q}(\mathcal{H})$  be the quotient map.

**Lemma 1** (Calkin obstruction). *Let  $A$  be a unital  $C^*$ -algebra, let  $\pi : A \rightarrow \mathcal{B}(\mathcal{H})$  be a  $*$ -representation, and let  $\psi : A \rightarrow \mathcal{B}(\mathcal{H})$  be a UCP map. If there is a unitary  $u \in \mathcal{B}(\mathcal{H})$  such that*

$$\pi(t) - u^* \psi(t) u \in \mathcal{K}(\mathcal{H}) \quad (t \in A),$$

*then  $q \circ \psi : A \rightarrow \mathcal{Q}(\mathcal{H})$  is a  $*$ -homomorphism.*

*Proof.* In the Calkin algebra,

$$q(\pi(t)) = q(u)^* q(\psi(t)) q(u) \quad (t \in A).$$

Hence

$$q(\psi(t)) = q(u) q(\pi(t)) q(u)^* \quad (t \in A).$$

The right-hand side is a unitary conjugate of the  $*$ -homomorphism  $q \circ \pi$ , so  $q \circ \psi$  is a  $*$ -homomorphism.  $\square$

**Theorem 1** (Amplified non-UEP obstruction). *Let  $\mathcal{M}$  be a separable unital operator space and let  $\pi_0 : C^*(\mathcal{M}) \rightarrow \mathcal{B}(\mathcal{H}_0)$  be a representation on a separable Hilbert space. Assume that  $\pi_0$  has AUEP with respect to  $\mathcal{M}$  but does not have UEP. Then there are a separable Hilbert space  $\mathcal{H}$ , a representation  $\Pi : C^*(\mathcal{M}) \rightarrow \mathcal{B}(\mathcal{H})$  with AUEP, and a UCP map  $\Psi : C^*(\mathcal{M}) \rightarrow \mathcal{B}(\mathcal{H})$  such that  $\Psi|_{\mathcal{M}} = \Pi|_{\mathcal{M}}$ , but no unitary  $u \in \mathcal{B}(\mathcal{H})$  satisfies*

$$\Pi(t) - u^* \Psi(t) u \in \mathcal{K}(\mathcal{H}) \quad (t \in C^*(\mathcal{M})).$$

*Consequently no AUEP-approximating net or sequence for this pair can have all of its pointwise errors compact.*

*Proof.* Because  $\pi_0$  is not UEP, there exists a UCP map  $\psi_0 : C^*(\mathcal{M}) \rightarrow \mathcal{B}(\mathcal{H}_0)$  such that  $\psi_0|_{\mathcal{M}} = \pi_0|_{\mathcal{M}}$  and  $\psi_0 \neq \pi_0$ . The map  $\psi_0$  cannot be multiplicative: if it were a  $*$ -homomorphism, then its agreement with  $\pi_0$  on the generating space  $\mathcal{M}$  would force  $\psi_0 = \pi_0$  on  $C^*(\mathcal{M})$ . Thus there exist  $a, b \in C^*(\mathcal{M})$  such that

$$D = \psi_0(ab) - \psi_0(a)\psi_0(b) \neq 0.$$

Let  $\mathcal{H} = \bigoplus_{n=1}^{\infty} \mathcal{H}_0$ , and define the countable amplifications

$$\Pi = \bigoplus_{n=1}^{\infty} \pi_0, \quad \Psi = \bigoplus_{n=1}^{\infty} \psi_0.$$

The Hilbert space  $\mathcal{H}$  is separable. The representation  $\Pi$  has AUEP by Thompson's direct-sum permanence theorem for AUEP [1, Proposition 3.7], and  $\Psi|_{\mathcal{M}} = \Pi|_{\mathcal{M}}$ .

The multiplicative defect of  $\Psi$  at  $a, b$  is

$$\Delta = \Psi(ab) - \Psi(a)\Psi(b) = \bigoplus_{n=1}^{\infty} D.$$

Since  $D \neq 0$ , choose  $\xi \in \mathcal{H}_0$  with  $\|D\xi\| > 0$ . The unit vectors placing  $\xi/\|\xi\|$  in the  $n$ th summand show that  $\Delta$  maps a bounded sequence to a sequence with no norm-convergent subsequence. Therefore  $\Delta$  is not compact.

If a unitary  $u$  made all errors  $\Pi(t) - u^*\Psi(t)u$  compact, then Lemma 1 would imply that  $q \circ \Psi$  is a  $*$ -homomorphism. In particular,

$$q(\Delta) = q(\Psi(ab) - \Psi(a)\Psi(b)) = 0.$$

That says exactly that  $\Delta$  is compact, contradicting the previous paragraph. Hence no such unitary exists.  $\square$

## 4 Concrete Counterexample from the Source Paper

**Corollary 1.** *The compact-refinement question raised in [1] has a negative answer even when both the operator space and the Hilbert space are separable.*

*Proof.* Let  $\mathcal{O}_\infty$  be the infinite Cuntz algebra with generators  $(V_n)_{n \in \mathbb{N}}$ , and let  $\mathcal{A}$  be the norm-closed unital operator algebra generated by these generators. Thompson recalls that  $C^*(\mathcal{A}) \cong \mathcal{O}_\infty$ , that all irreducible representations of  $\mathcal{O}_\infty$  are approximately unitarily equivalent because  $\mathcal{O}_\infty$  is simple, and that there is an irreducible representation  $\rho$  which is not a boundary representation for  $\mathcal{A}$  [1, Example 2]. Since  $\rho$  is not a boundary representation, it does not have UEP with respect to  $\mathcal{A}$ .

The same example supplies boundary representations, and all irreducible representations are approximately unitarily equivalent. Therefore  $\rho$  is approximately unitarily equivalent to a representation with UEP. By Thompson’s Theorem 3.6,  $\rho$  has AUEP. Because  $\mathcal{O}_\infty$  is separable and  $\rho$  is irreducible,  $\rho$  acts on a separable Hilbert space: any nonzero vector is cyclic and the orbit of a countable dense subset of  $\mathcal{O}_\infty$  is dense.

Apply Theorem 1 to  $\mathcal{M} = \mathcal{A}$  and  $\pi_0 = \rho$ . The resulting countable amplification is a separable AUEP representation with a UCP extension for which compact pointwise errors are impossible. This is precisely a negative instance of the compact-refinement property under separability assumptions.  $\square$

## 5 Verification Notes and Limitations

The proof is purely structural; no numerical computation is used. The main check is that compact pointwise errors imply multiplicativity in the Calkin quotient, whereas countable amplification turns any nonzero multiplicative defect into a noncompact diagonal defect.

The result answers the general compact-refinement question negatively. It does not rule out positive refinements under additional hypotheses, such as UEP itself, postliminal envelope assumptions, or restrictions on the Calkin image of the chosen UCP extension.

## 6 Novelty Check

The run’s lightweight indexes were searched for the arXiv id, the exact title, “approximate unique extension property”, “compact refinement”, “Calkin”, and the Voiculescu compact-error phrase. No previous packet for this exact target was found. A web search on 2026-07-03 for exact close variants of “approximate unique extension property” with “compact”, “Calkin”, and “Voiculescu” found the source arXiv paper but no separate resolution. Novelty confidence is therefore bounded-local-plus-web rather than exhaustive.

## 7 Human Review Recommendation

Review as a candidate counterexample. The most important points to check are: (1) Thompson's Example 2 really supplies a separable AUEP representation that is not UEP; (2) the non-UEP extension is necessarily nonmultiplicative; and (3) the Calkin-quotient argument forbids even one compact-error conjugating unitary.

## References

- [1] Ian Thompson. *An approximate unique extension property for completely positive maps*. arXiv:2301.08285v1, 2023.